EXACT NON-SCATTERING HOMOGENEOUS DIELECTRICS FROM SCHIFFER COUNTEREXAMPLES

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Abstract. We prove the existence of bounded, simply connected, noncircular homogeneous isotropic dielectric inclusions that are exactly non-scattering for a nonzero physical incident field in the two-dimensional scalar Maxwell reduction. The incident field is the explicit normalized Herglotz wave

$$v(x) = \sqrt{2\pi}\, J_0(|x|), \qquad h(d) = \frac{1}{\sqrt{2\pi}},$$

at exterior wavenumber one. The construction starts from the computer-assisted noncircular Schiffer counterexample of Colbrook and Stepaniants. We replace its constant boundary datum by a small-frequency Bessel wave, prove that the regular Schiffer zero persists, and certify the continuation uniformly for every $0 < \tau \leq 10^{-13}$. After dilation, the resulting domain and constant relative permittivity are

$$\Omega_\tau = \sqrt{\tau}\, \phi_{p_\tau}(\mathbb{D}), \qquad q_\tau = \frac{p_{\tau,0}^2}{\tau}.$$

An interior Helmholtz field matches both Cauchy traces of v exactly, so the radiating scattered field and its far-field pattern vanish identically. As a consequence, every prescribed contrast $Q \geq 1.03 \times 10^{16}$ occurs for some noncircular member of the family.

1. Introduction

Let $D \subset \mathbb{R}^2$ be a bounded dielectric inclusion with constant relative permittivity $q > 0$ and constant magnetic permeability. In the out-of-plane-electric scalar reduction of time-harmonic Maxwell equations, an incident field v at exterior wavenumber $k > 0$ is physical when it is a nonzero entire solution of

$$(\Delta + k^2)v = 0 \qquad \text{in } \mathbb{R}^2. \tag{1}$$

Exact non-scattering requires an interior field U satisfying

$$(\Delta + k^2 q)U = 0 \qquad \text{in } D \tag{2}$$

and the two trace identities

$$U = v, \qquad \partial_\nu U = \partial_\nu v \qquad \text{on } \partial D. \tag{3}$$

Then the exterior total field may equal v itself, so the scattered field is the zero radiating solution. This is stronger than the mere existence of a transmission eigenvalue: the incident component must extend to an entire physical Helmholtz solution, and in the present theorem it is an L^2-Herglotz wave. The distinction is emphasized in [HV25, CV26].

Two lines of prior work frame this question. First, geometric obstructions: domains with corners scatter every incident wave [BPS14, PSV17, EH15], and non-scattering forces strong boundary regularity [SS21, CV23, CVX23], so any example must have smooth, indeed essentially

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real-analytic, boundary. Second, spectral necessity: non-scattering requires an interior transmission eigenpair [Kir86, CM88, CCH16], but a transmission eigenpair alone does not supply a physical incident field, and for broad families of perturbed discs the nonscattering wave numbers of physical incident waves are at most finite [VX21, HV25]. Radial homogeneous inclusions are the classical exception; whether any noncircular homogeneous inclusion admits an exactly non-scattering physical incident wave has remained open [CV26].

A neighboring overdetermined problem is Schiffer's conjecture, which is closely tied to the Pompeiu problem [Pom29, BST73, Wil76, Ber80]. It asks whether a bounded domain admitting a nonconstant solution of

$$(\Delta + \lambda)u = 0 \quad \text{in } D, \qquad u = 1, \quad \partial_\nu u = 0 \quad \text{on } \partial D \tag{4}$$

must be a disc. Colbrook and Stepaniants recently constructed a computer-assisted noncircular counterexample with real-analytic boundary, and Cao-Labora and de Dios Pont independently produced infinitely many bifurcating counterexamples [CS26, CDP26]. These results do not immediately solve the scattering problem: at positive frequency the constant function 1 does not satisfy the exterior Helmholtz equation.

The central observation of this paper is that the Schiffer datum can be replaced by a genuine radial Helmholtz wave. More precisely, a regular Schiffer zero persists when the constant boundary datum is deformed to

$$J_0(\sqrt{\tau}\,|x|) = 1 - \frac{\tau}{4}\,|x|^2 + O(\tau^2).$$

The resulting transmission pair has exterior squared wavenumber τ and interior squared wavenumber close to the Schiffer eigenvalue. A final dilation fixes the exterior wavenumber at $k = 1$. This answers the constant-index, noncircular, physical-incident-field existence question affirmatively in the two-dimensional scalar model. Thus the construction is a subwavelength, high-contrast resonance: the physical diameter shrinks like $\sqrt{\tau}$, while the interior optical size remains finite.

1.1. **Main result.** Let

$$h(d) = \frac{1}{\sqrt{2\pi}}, \qquad v(x) = \int_{\mathbb{S}^1} e^{ix\cdot d} h(d)\,\mathrm{d}s(d) = \sqrt{2\pi}\,J_0(|x|). \tag{5}$$

Then $h \in L^2(\mathbb{S}^1)$, $\|h\|_{L^2} = 1$, and $(\Delta + 1)v = 0$ in $\mathbb{R}^2$.

Theorem 1.1 (Exact non-scattering family). *For every $0 < \tau \le 10^{-13}$ there exist a bounded simply connected domain $\Omega_\tau \subset \mathbb{R}^2$, a constant $q_\tau > 0$, and an interior field U_τ such that:*

(i) *$\partial\Omega_\tau$ is a real-analytic Jordan curve and Ω_τ is not a disc;*
(ii) *$q_\tau \neq 1$;*
(iii)
$$(\Delta + q_\tau)U_\tau = 0 \quad \text{in } \Omega_\tau;$$
(iv) *$U_\tau = v$ and $\partial_\nu U_\tau = \partial_\nu v$ on $\partial\Omega_\tau$.*

Consequently the radiating scattered field and the far-field pattern vanish identically for the normalized Herglotz incident wave (5) at exterior wavenumber $k = 1$.

The construction is exact, not a finite multipole cancellation. For each τ, a contraction theorem selects a unique infinite coefficient pair (γ_τ, p_τ) in a fixed certified Banach ball. If

$$\phi_p(z) = z + \sum_{j\ge 1} \frac{p_j}{p_0(10j+1)} z^{10j+1}, \tag{6}$$

then

$$\Omega_\tau = \sqrt{\tau}\,\phi_{p_\tau}(\mathbb{D}), \qquad q_\tau = \frac{p_{\tau,0}^2}{\tau}. \tag{7}$$

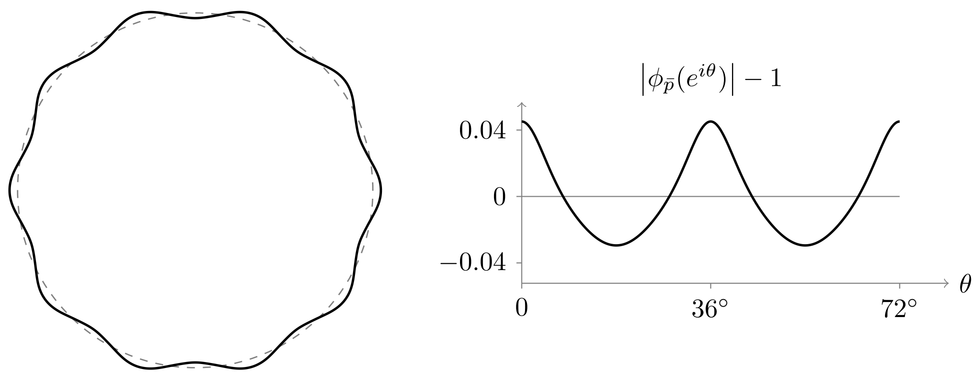

FIGURE 1. The certified Schiffer shape. Left: the image of the unit circle under $\phi_{\bar{p}}$ for the exact certificate center $\bar{p}$ of [CS26, CS26c] (solid), against the unit circle (dashed); the boundary radius varies between 0.9705 and 1.0452. Right: the radial profile $\left|\phi_{\bar{p}}(e^{i\theta})\right| - 1$ over two of the ten symmetry periods. Every domain $\Omega_\tau = \sqrt{\tau}\,\phi_{p_\tau}(\mathbb{D})$ of theorem 1.1 is a dilate of a small perturbation of this shape: the certified zero stays within 10^{-6} of $\bar{x}$ in the weighted norm.

Corollary 1.2 (Prescribed high contrast)**.** *For every real $Q \geq 1.03 \times 10^{16}$ there exists a bounded simply connected non-disc with real-analytic Jordan boundary and constant relative permittivity exactly Q that is non-scattering for the same incident wave* (5) *at exterior wavenumber one.*

1.2. **Proof mechanism.** The argument has four steps.

(1) The Colbrook–Stepaniants theorem supplies a regular noncircular Schiffer zero in a weighted coefficient algebra.
(2) We introduce a Bessel-deformed fixed-disc equation $F_\tau = 0$ whose zero is exactly equivalent to matching the physical Bessel Cauchy data.
(3) The implicit-function theorem gives a local branch abstractly, and explicit perturbation bounds show that the original contraction ball remains valid for all $0 \leq \tau \leq 10^{-13}$.
(4) Conformal covariance and dilation produce the physical domain, constant contrast, and exact zero scattered field.

The full numerical provenance, the executable verifiers, and a Lean re-verification of the certificate arithmetic are separated into the accompanying verification supplement; the present article contains the mathematical construction and proof.

2. The certified Schiffer seed

We use the fixed-disc formulation of [CS26]. Let X_ρ and Y_ρ be their weighted real coefficient spaces, with $\rho = 1.05$. A shape sequence has ten-fold symmetry,

$$p(z) = \sum_{j \geq 0} p_j z^{10j}, \qquad p_0 > 0, \tag{8}$$

and is related to (6) by

$$p = p_0 \phi_p'. \tag{9}$$

The operator K is an explicit inverse of the Laplacian on the compatible coefficient range and satisfies

$$\Delta K\gamma = \gamma, \qquad K\gamma = \partial_r K\gamma = 0 \quad \text{on } \partial\mathbb{D}. \tag{10}$$

The Schiffer equation is

$$F_0(\gamma, p) := \gamma + |p|^2 (1 + K\gamma) = 0. \tag{11}$$

Indeed, if $W = K\gamma$ and $\widetilde{u} = 1 + W$, then

$$\Delta\widetilde{u} + |p|^2\,\widetilde{u} = 0 \quad \text{in } \mathbb{D}, \qquad \widetilde{u} = 1, \quad \partial_r\widetilde{u} = 0 \quad \text{on } \partial\mathbb{D}.$$

Using (9), the push-forward to $\phi_p(\mathbb{D})$ has constant interior squared wavenumber p_0^2.

The following summarizes precisely the consequences of the Colbrook–Stepaniants computer-assisted theorem used here.

Proposition 2.1 (Certified Schiffer seed)**.** *There is a center $\bar{x} = (\bar{\gamma}, \bar{p}) \in X_\rho$, a radius $r = 10^{-6}$, an invertible bounded operator $\mathcal{A} : Y_\rho \to X_\rho$, and a unique exact zero $x_\star = (\gamma_\star, p_\star)$ of F_0 in the closed ball $B_r(\bar{x})$ with the following properties.*

(a) *Let $T_0 = I - \mathcal{A}F_0$. The certificate provides exact cubic Taylor majorants $Y \le 1.59 \times 10^{-10}$, $Z \le 0.621$, $C_2 \le 122$, $C_3 \le 0.012$ such that, on $B_r(\bar{x})$,*

$$\sup_{B_r(\bar{x})} \|T_0(x) - \bar{x}\| \le Y + Zr + C_2 r^2 + C_3 r^3, \qquad \sup_{B_r(\bar{x})} \|DT_0\| \le Z + 2C_2 r + 3C_3 r^2.$$

Writing $R_0 := Y + Zr + C_2r^2 + C_3r^3 - r$ for the invariant-ball defect and $L_0 := Z + 2C_2r + 3C_3r^2$ for the contraction constant, exact rational evaluation gives

$$R_0 \le -0.000000378718999999988 < 0, \tag{12}$$

$$L_0 \le 0.621244000000036 < 1, \tag{13}$$

so T_0 maps $B_r(\bar{x})$ into itself and is a strict contraction there.

(b) *Every p in the ball satisfies*

$$\|p\|_\rho < 55, \qquad 31 < p_0 < 32, \qquad \sum_{j \ge 1} |p_j/p_0| < 0.65. \tag{14}$$

(c) *ϕ_p is univalent on a neighborhood of $\overline{\mathbb{D}}$, and its first nonconstant shape coefficient does not vanish. Hence $\phi_p(\mathbb{D})$ is a bounded simply connected non-disc with real-analytic Jordan boundary.*

(d) *The preconditioner satisfies*

$$\|\mathcal{A}\| < 1180. \tag{15}$$

Proposition 2.1 restates, in the notation of this paper, the certified outputs of [CS26, Eqs. (3)–(5) and (22), Lemma 2.4, Theorem 1.1] and of the released validation certificate [CS26c], whose pinned hashes and replay commands are recorded in the verification supplement. The bound (15) follows from the upstream identity $C_3 = \|\mathcal{A}\| / (96\bar{p}_0^2)$ together with $C_3 \le 0.012$ and the exact center value $\bar{p}_0 < 32$:

$$\|\mathcal{A}\| = 96\bar{p}_0^2 C_3 < 96 \cdot 32^2 \cdot 0.012 = 1179.648 < 1180.$$

It therefore does not depend on reconstructing the norm of the upstream approximate inverse to many digits.

Remark 2.2 (Epistemic status)**.** *Theorem 1.1 is conditional on the computer-assisted theorem of* [CS26]*, which at the time of writing is a very recent preprint. Everything imported from it is confined to proposition 2.1; the accompanying release pins the upstream certificate by cryptographic hash, replays its authenticated checker, and machine-compares every constant in proposition 2.1 against the pinned upstream data.*

Lemma 2.3 (Regularity of the certified zero)**.** *The derivative $DF_0(x_\star) : X_\rho \to Y_\rho$ is a bounded linear isomorphism.*

Proof. At the fixed point,
$$DT_0(x_\star) = I - \mathcal{A}DF_0(x_\star).$$
By (13), $\|DT_0(x_\star)\| < 1$. Hence $I - DT_0(x_\star) = \mathcal{A}DF_0(x_\star)$ is invertible by the Neumann-series lemma. Since $\mathcal{A}$ is invertible, so is $DF_0(x_\star)$. ◻

3. The Bessel-deformed transmission equation

For $\tau \geq 0$, define

$$H_{\tau,p}(z) := \sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)^2} \left(\frac{\tau\, |\phi_p(z)|^2}{4} \right)^n . \tag{16}$$

For $\tau > 0$,

$$H_{\tau,p}(z) = J_0\big(\sqrt{\tau}\ |\phi_p(z)|\big)\,, \tag{17}$$

and $H_{0,p} = 1$. The power-series definition makes the dependence on τ analytic at zero. Conformal covariance of the Laplacian and (9) give

$$\Delta H_{\tau,p} + \frac{\tau}{p_0^2}\, |p|^2\, H_{\tau,p} = 0 \quad \text{in } \mathbb{D}. \tag{18}$$

We now define the deformed operator

$$F_\tau(\gamma, p) := \gamma + |p|^2\, K\gamma + |p|^2 \left(1 - \frac{\tau}{p_0^2}\right) H_{\tau,p}. \tag{19}$$

At $\tau = 0$, this is exactly (11).

Lemma 3.1 (Fixed-disc transmission identity)**.** *If $F_\tau(\gamma, p) = 0$ and*

$$\widetilde{U}_{\tau,p} := H_{\tau,p} + K\gamma, \tag{20}$$

then

$$\Delta \widetilde{U}_{\tau,p} + |p|^2\, \widetilde{U}_{\tau,p} = 0 \quad \textit{in } \mathbb{D}. \tag{21}$$

Moreover, $\widetilde{U}_{\tau,p}$ and $H_{\tau,p}$ have identical Dirichlet and radial Neumann traces on $\partial\mathbb{D}$.

Proof. Using (10) and (18), the left-hand side of (21) is exactly $F_\tau(\gamma, p)$. The trace statement follows from the two clamped trace identities in (10). ◻

The preceding equation reveals the continuation mechanism independently of the particular numerical seed.

Theorem 3.2 (Regular Schiffer zeros generate physical non-scatterers)**.** *Let $x_\star = (\gamma_\star, p_\star)$ be a zero of F_0 such that $DF_0(x_\star)$ is invertible, $p_{\star,0} > 0$, and $\phi_{p_\star}$ is univalent on a neighborhood of $\overline{\mathbb{D}}$ and non-affine. Then there are $\varepsilon > 0$ and a real-analytic branch*

$$(-\varepsilon, \varepsilon) \ni \tau \longmapsto x_\tau = (\gamma_\tau, p_\tau)$$

with $F_\tau(x_\tau) = 0$ and $x_\tau \to x_\star$ as $\tau \to 0$. For every sufficiently small $\tau > 0$, the scaled domain $\sqrt{\tau}\, \phi_{p_\tau}(\mathbb{D})$ is a noncircular homogeneous dielectric that is exactly non-scattering for the incident wave (5) at exterior wavenumber one.

Proof. The series (16) converges normally in a neighborhood of $(0, x_\star)$ in the weighted coefficient algebra. Hence $(\tau, x) \mapsto F_\tau(x)$ is real analytic there. The analytic implicit-function theorem and the invertibility of $DF_0(x_\star)$ produce the branch. Univalence on a complex collar and non-affineness are open in the weighted coefficient norm, so they persist for small τ. Non-affineness excludes a disc through the ten-fold symmetry of the ansatz (6): with $\omega = e^{2\pi i/10}$ one has $\phi_p(\omega z) = \omega \phi_p(z)$, so the image is invariant under rotation by ω; a disc with this invariance is centered at the origin, and the unique Riemann map of $\mathbb{D}$ onto such a disc with $\phi(0) = 0$ and $\phi'(0) > 0$ is linear. A non-affine ϕ_p therefore cannot map $\mathbb{D}$ onto a disc.

For $\tau > 0$, lemma 3.1 supplies two fields with equal Cauchy data on the unit circle. Pushing them forward by ϕ_{p_τ} gives exterior squared wavenumber τ and interior squared wavenumber $p_{\tau,0}^2$. Dilation by $\sqrt{\tau}$ fixes the exterior wavenumber at one and changes the interior squared wavenumber to $p_{\tau,0}^2/\tau$. The complete physical verification is given in section 5. □

4. A uniform certified branch

The abstract theorem gives a local branch but no explicit interval. We now show that the entire Schiffer contraction ball remains valid for $0 \le \tau \le 10^{-13}$. Put

$$\delta F_\tau := F_\tau - F_0.$$

All norms below are the weighted coefficient majorants of [CS26]. One structural property of that algebra is used repeatedly and deserves to be isolated.

Lemma 4.1 (Radial multiplication is nonexpansive)**.** *In the weighted coefficient algebra of* [CS26, CS26c] *the norm is a weighted ℓ^1 sum over a basis indexed by an angular block index ℓ and a radial index s, with weight $c_\ell \rho^\ell$ depending on ℓ only. Multiplication by the radial factor $|z|^2$ raises the radial index and leaves the angular index and the weight unchanged. Hence $\left\| |z|^2 f \right\| \le \|f\|$ for every f, and for any product decomposition $\phi_p(z) = z\psi_p(z)$,*

$$\left\| |\phi_p|^2 \right\| = \left\| |z|^2 \, |\psi_p|^2 \right\| \le \|\psi_p\|^2 .$$

Proof. The norm and the multiplication recurrences are specified in [CS26c, Validation Specification, §§2 and 4]: the weights are $c_\ell \rho^\ell$ with $c_0 = 1$, $c_\ell = 2$, independent of the radial index, and multiplication by r^2 acts as the pure radial shift $(\ell, s) \mapsto (\ell, s+1)$. A shift with unchanged weights does not increase a weighted ℓ^1 norm, and the final inequality is submultiplicativity of the algebra norm. □

Write $\phi_p(z) = z\psi_p(z)$. From (6) and (14),

$$\|\psi_p\| \le 1 + \frac{1}{11p_0} \sum_{j \ge 1} |p_j| \, \rho^j \le 1 + \frac{\|p\|_\rho - p_0}{11 \cdot 31} \le 1 + \frac{55 - 31}{11 \cdot 31} < \frac{11}{10}, \tag{22}$$

and therefore, by lemma 4.1,

$$\left\| |\phi_p|^2 \right\| \le \|\psi_p\|^2 < \frac{121}{100} < \frac{5}{4}. \tag{23}$$

Set $u = \tau \left\| |\phi_p|^2 \right\| /4$. For $0 \le \tau \le 10^{-13}$, $u < 1/2$. The full series (16), without truncation, therefore gives

$$\|H_{\tau,p}\| \le 2, \qquad \|H_{\tau,p} - 1\| \le \frac{5}{8}\tau. \tag{24}$$

Since $p_0 > 31$,

$$\begin{aligned} \left\| \left(1 - \frac{\tau}{p_0^2}\right) H_{\tau,p} - 1 \right\| &\le \|H_{\tau,p} - 1\| + \frac{\tau}{p_0^2} \|H_{\tau,p}\| \\ &< \frac{63}{100}\tau. \end{aligned} \tag{25}$$

Multiplication by $\|p\|^2 < 55^2$ yields

$$\boxed{\|\delta F_\tau\| < 1906\tau.} \tag{26}$$

For the derivative estimate, let ξ be a shape variation of full X_ρ norm at most one. The upstream product norm weights the shape block by $2\bar{p}_0\rho^j$ [CS26c, Validation Specification, Eq. (13)], so

$$\|\xi\| \le \frac{1}{2\bar{p}_0} < \frac{1}{62}, \qquad |\xi_0| < \frac{1}{62}. \tag{27}$$

Differentiating (6) and using (14) gives

$$\|D\psi_p[\xi]\| \le \frac{1}{11}\left(\frac{\|\xi\|}{31} + \frac{55\,|\xi_0|}{31^2}\right) < \frac{1}{7000}. \tag{28}$$

Consequently

$$\left\|D\,|\phi_p|^2\,[\xi]\right\| \le 2\,\|\psi_p\|\,\|D\psi_p[\xi]\| < \frac{1}{3000}, \qquad \|DH_{\tau,p}[\xi]\| \le \frac{\tau}{3000}, \tag{29}$$

where the first estimate again uses lemma 4.1. Differentiating (19), and combining (25), (24), (27), and (29), gives

$$\boxed{\|D\delta F_\tau\| < 3\tau.} \tag{30}$$

The elementary constant arithmetic behind (26) and (30) is recorded in the verification supplement.

Define

$$T_\tau(x) = x - \mathcal{A}F_\tau(x).$$

On the original radius-r ball, (15), (26), and (30) give

$$R_\tau \le R_0 + 1180 \cdot 1906\,\tau, \tag{31}$$

$$L_\tau \le L_0 + 1180 \cdot 3\,\tau. \tag{32}$$

At $\tau_* = 10^{-13}$,

$$R_{\tau_*} \le -0.000000153810999999988 < 0, \tag{33}$$

$$L_{\tau_*} \le 0.621244000354036 < 1. \tag{34}$$

The right-hand sides of (31)–(32) increase with τ. Hence T_τ maps the same closed ball into itself and is a strict contraction for every $0 \le \tau \le \tau_*$. Banach's theorem gives a unique exact zero

$$x_\tau = (\gamma_\tau, p_\tau) \in B_r(\bar{x}) \qquad (0 \le \tau \le 10^{-13}). \tag{35}$$

Because the whole branch stays in the certified ball, all univalence, analytic-boundary, and noncircularity conclusions of proposition 2.1 hold uniformly.

5. The physical inclusion and exact zero scattering

Fix $0 < \tau \le 10^{-13}$ and the exact zero $x_\tau = (\gamma_\tau, p_\tau)$ from (35). Define the domain and contrast by (7). For $y \in \Omega_\tau$, put

$$z = \phi_{p_\tau}^{-1}\left(\frac{y}{\sqrt{\tau}}\right)$$

and define

$$U_\tau(y) = \sqrt{2\pi}\left[H_{\tau,p_\tau}(z) + K\gamma_\tau(z)\right]. \tag{36}$$

Since $y = \sqrt{\tau}\,\phi_{p_\tau}(z)$,

$$H_{\tau,p_\tau}(z) = J_0(|y|). \tag{37}$$

Conformal covariance transforms (21) into an interior Helmholtz equation with squared wavenumber $p_{\tau,0}^2$ on $\phi_{p_\tau}(\mathbb{D})$. The dilation $y = \sqrt{\tau}\, x$ therefore gives

$$(\Delta_y + q_\tau)U_\tau = 0 \quad \text{in } \Omega_\tau, \qquad q_\tau = \frac{p_{\tau,0}^2}{\tau}. \tag{38}$$

By (14),

$$q_\tau > \frac{31^2}{10^{-13}} > 1, \tag{39}$$

so q_τ is positive and nonunit.

It remains to check the traces. The difference between the pulled-back interior field and the pulled-back incident field is $K\gamma_\tau$, whose Dirichlet and radial Neumann traces vanish by (10). Under a conformal map the normal derivative is multiplied by a nonzero boundary scale factor, and under dilation it is multiplied by another nonzero constant. Thus zero remains zero, and (37) gives

$$U_\tau = v, \qquad \partial_\nu U_\tau = \partial_\nu v \quad \text{on } \partial\Omega_\tau. \tag{40}$$

For completeness, writing $d = (\cos\theta, \sin\theta)$,

$$\int_0^{2\pi} e^{ix\cdot d}\, \mathrm{d}\theta = 2\pi J_0(|x|),$$

so (5) follows, and

$$\|h\|_{L^2(\mathbb{S}^1)}^2 = \int_0^{2\pi} \frac{1}{2\pi}\, \mathrm{d}\theta = 1.$$

The incident field is therefore a nonzero normalized L^2-Herglotz wave.

Define the total field piecewise by U_τ in Ω_τ and by v in the exterior. Equation (40) makes it a transmission solution. Its exterior scattered component is identically zero, which satisfies the Sommerfeld radiation condition and has zero far-field pattern. Because q_τ is a bounded positive constant, the direct transmission problem for the incident field v has exactly one solution [CK19, Chapter 8]; the solution just exhibited is therefore the physical total field, and the scattered field of Ω_τ for v vanishes identically. This proves theorem 1.1.

6. Consequences and interpretation

6.1. **Prescribing the contrast.** The contraction maps T_τ depend continuously on τ and have a uniform Lipschitz constant strictly below one. Hence their unique fixed points depend continuously on τ. One may see this directly from

$$\|x_\tau - x_\sigma\| \le \frac{\|T_\tau(x_\sigma) - T_\sigma(x_\sigma)\|}{1 - L}, \tag{41}$$

where $L < 1$ is any common contraction constant. Thus $\tau \mapsto p_{\tau,0}$ and

$$q_\tau = \frac{p_{\tau,0}^2}{\tau}$$

are continuous on $(0, 10^{-13}]$. Since $p_{\tau,0} \to p_{\star,0} > 31$ as $\tau \downarrow 0$,

$$q_\tau \longrightarrow +\infty. \tag{42}$$

At $\tau_* = 10^{-13}$, the upper bound $p_{\tau_*,0} < 32$ gives

$$q_{\tau_*} < \frac{32^2}{10^{-13}} = 1.024 \times 10^{16}. \tag{43}$$

For any $Q \geq 1.03 \times 10^{16}$, choose $\delta > 0$ small enough that $q_\delta > Q$. The intermediate-value theorem on $[\delta, \tau_*]$ and (43) produce some τ with $q_\tau = Q$. This proves corollary 1.2.

6.2. **The subwavelength/high-index regime.** At fixed exterior wavenumber one, the diameter of Ω_τ is $O(\sqrt{\tau})$, whereas

$$\sqrt{q_\tau}\,\sqrt{\tau} = p_{\tau,0} \longrightarrow p_{\star,0} \approx 31.967. \tag{44}$$

Thus the family maintains a finite interior optical size by increasing the positive refractive index as the physical inclusion shrinks. The result is an existence theorem for homogeneous, passive, positive-index material, but it does not yet reach moderate contrast. Natural next questions are whether other Schiffer branches, nonradial Herglotz controls, or continuation farther from $\tau = 0$ can reduce the required contrast, and whether an analogous mechanism exists for the full three-dimensional vector Maxwell system.

Reproducibility and formal verification

The accompanying release contains the exact certificate data, independent exact-arithmetic verifiers that rederive the perturbation constants of section 4 from the certified ball bounds, the pinned and replayed upstream dependency, and a Lean 4 re-verification of the exact rational certificate arithmetic together with generic fixed-point lemmas in the form used here. The Lean development does not formalize the operator estimates, the upstream certificate, or the scattering analysis; its exact boundary, the numerical provenance, and the replay commands are recorded in a separate verification supplement so that they do not obscure the mathematical argument above.

Declaration of computational assistance

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References

[Ber80] Carlos A. Berenstein. An inverse spectral theorem and its relation to the Pompeiu problem. *J. Analyse Math.* 37 (1980), 128–144.

[BPS14] Eemeli Blåsten, Lassi Päivärinta, and John Sylvester. Corners always scatter. *Comm. Math. Phys.* 331 (2014), no. 2, 725–753.

[BST73] Leon Brown, Bertram M. Schreiber, and B. Alan Taylor. Spectral synthesis and the Pompeiu problem. *Ann. Inst. Fourier (Grenoble)* 23 (1973), no. 3, 125–154.

[CCH16] Fioralba Cakoni, David Colton, and Houssem Haddar. *Inverse Scattering Theory and Transmission Eigenvalues.* CBMS-NSF Regional Conference Series in Applied Mathematics 88, SIAM, Philadelphia, 2016.

[CV23] Fioralba Cakoni and Michael S. Vogelius. Singularities almost always scatter: regularity results for non-scattering inhomogeneities. *Comm. Pure Appl. Math.* 76 (2023), no. 12, 4022–4047.

[CV26] Fioralba Cakoni and Michael S. Vogelius. Transmission eigenvalues and non-scattering. arXiv:2602.06250, 2026.

[CVX23] Fioralba Cakoni, Michael S. Vogelius, and Jingni Xiao. On the regularity of non-scattering anisotropic inhomogeneities. *Arch. Ration. Mech. Anal.* (2023), doi:10.1007/s00205-023-01863-y.

[CDP26] Gonzalo Cao-Labora and Jaume de Dios Pont. Counterexamples to Schiffer's conjecture. arXiv:2608.05114, 2026.

[CS26] Matthew J. Colbrook and George Stepaniants. A computer-assisted counterexample to the planar Pompeiu and Schiffer conjectures. arXiv:2608.01579, 2026.

[CS26c] Matthew J. Colbrook and George Stepaniants. Pompeiu–Schiffer validation certificate v1.0.0. Zenodo, 2026, doi:10.5281/zenodo.21765287.

[CK19] David Colton and Rainer Kress. *Inverse Acoustic and Electromagnetic Scattering Theory.* 4th edition, Applied Mathematical Sciences 93, Springer, Cham, 2019.

[CM88] David Colton and Peter Monk. The inverse scattering problem for time-harmonic acoustic waves in an inhomogeneous medium. *Quart. J. Mech. Appl. Math.* 41 (1988), no. 1, 97–125.

[EH15] Johannes Elschner and Guanghui Hu. Corners and edges always scatter. *Inverse Problems* 31 (2015), no. 1, 015003.

[HV25] Narek Hovsepyan and Michael S. Vogelius. Scattering from analytic and piecewise analytic inhomogeneities. arXiv:2507.13986, 2025.

[Kir86] Andreas Kirsch. The denseness of the far field patterns for the transmission problem. *IMA J. Appl. Math.* 37 (1986), no. 3, 213–225.

[PSV17] Lassi Päivärinta, Mikko Salo, and Esa V. Vesalainen. Strictly convex corners scatter. *Rev. Mat. Iberoam.* 33 (2017), no. 4, 1369–1396.

[Pom29] Dimitrie Pompeiu. Sur certains systèmes d'équations linéaires et sur une propriété intégrale des fonctions de plusieurs variables. *C. R. Acad. Sci. Paris* 188 (1929), 1138–1139.

[SS21] Mikko Salo and Henrik Shahgholian. Free boundary methods and non-scattering phenomena. *Res. Math. Sci.* 8 (2021), no. 4, Paper No. 58.

[VX21] Michael S. Vogelius and Jingni Xiao. Finiteness results concerning nonscattering wave numbers for incident plane and Herglotz waves. *SIAM J. Math. Anal.* 53 (2021), no. 5, doi:10.1137/20M1367854.

[Wil76] Stephen A. Williams. A partial solution of the Pompeiu problem. *Math. Ann.* 223 (1976), no. 2, 183–190.

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